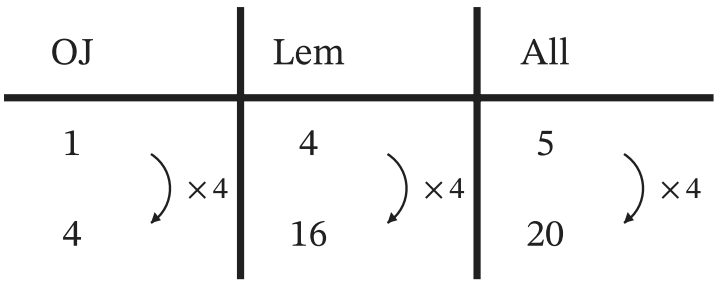
13 Ben fills a glass with orange juice and lemonade in the ratio 1 : 4 by volume.
He mixes the liquid that is in the glass.

Ben drinks $\frac{1}{4}$ of this liquid.

He then fills the glass using orange juice.

Work out the ratio of orange juice to lemonade, by volume, that is now in the glass.
Give your ratio in its simplest form.

1 mark
Choose volumes in ratio 1:4.



Take a full glass as 20 equal units.

1 mark
Remove ¼ of both liquids.

Drink ¼: orange 4 − 1 = 3; lemonade 16 − 4 = 12.

Add 5 orange: 3 + 5 = 8; lemonade stays 12.

1 mark
Refill with juice then simplify.

Orange : lemonade = 8 : 12 = 2 : 3

(Total for Question 13 is 3 marks)